```
import numpy as np
np.random.seed(42)
x=np.random.rand(100,2) @ np.array([[2,1],[1,3]]) # @ sign is used for multiplication
print(x)
```

```
 [0.94714797 1.28653229]
 [2.09676983 2.64227859]
 [2.24664041 2.30385752]
 [0.95243328 2.25932861]
 [2.08284729 2.44461664]
 [2.03572996 2.25235397]
 [1.47300668 1.80535588]
 [0.15872968 0.34909341]
 [0.69926878 1.94066042]
 [1.13728265 1.84006805]
 [2.06442518 1.65544316]
 [1.57631698 2.67703634]
 [0.53457624 0.45973789]
 [0.74072419 0.77341531]
 [2.66751568 3.35405879]
 [2.1382681  3.24778553]
 [1.79391421 1.36338225]
 [2.32446024 2.51058572]
 [2.51097161 3.49571405]
 [0.74605887 0.64815925]
 [0.88297811 1.50925853]
 [2.49676012 3.40020652]
 [0.52465156 1.53919404]
 [1.05692982 1.08373443]
 [0.57734591 1.13271088]
 [2.20902234 1.9125185 ]
 [1.7406002  2.6278475 ]
 [1.69904129 3.27897585]
 [2.17667689 1.71779418]
 [1.29537532 1.39988344]
 [0.60656794 0.39550134]
 [1.72180769 2.1176014 ]
 [0.38160397 0.88741814]
 [2.05609366 1.62695156]
 [0.7792425  1.61325315]
 [2.21335618 1.71181627]
 [2.10589071 2.95699439]
 [1.20349144 2.42228659]
 [1.3678721  2.26470062]
 [1.80283411 2.24085376]
 [1.01588204 2.59619726]
 [0.82807864 0.8803356 ]
 [0.67244323 1.81345397]
 [1.37171655 0.72732785]
 [1.25068189 1.19158038]
 [1.46471201 1.16827208]
 [1.76861082 1.85114378]
 [2.01098092 1.34929282]
 [0.79560622 0.68148691]
 [2.72672659 3.55671168]
 [1.1758673  2.23789377]
 [2.18964521 2.48282463]
 [1.30115345 1.25520745]
 [1.08342129 2.78475004]
 [2.43393757 2.79972243]
 [1.02726916 1.38665851]
 [2.34902162 3.41728646]
 [2.55404839 3.22671306]]
```

```
# step 2 : Standardize
x_mean = x - np.mean(x,axis=0)

# step 3 : Covariance matrix
cov_matrix = np.cov(x_mean, rowvar=False)
print("Covariance matrix: ",cov_matrix)

# STEP 4 : Eigenvalues and Eigenvectors
eigenvalues, eigenvectors = np.linalg.eigh(cov_matrix)
print("Eigenvalues: ",eigenvalues)
print("Eigenvectors: ",eigenvectors)

#STEP 5 : sort eigenvectors by eigenvalues (descending)
sorted_idx = np.argsort(eigenvalues)[::-1]

eigenvalues = eigenvalues[sorted_idx]
eigenvectors = eigenvectors[:,sorted_idx]
print("Eigenvalues: ",eigenvalues)
print("Eigenvectors: ",eigenvectors)

#step 6 : project data onto first principal component (reduce to 1D)
```

```
pc1=eigenvectors[:,0]
print("pc1 \n",pc1)

x_pca=x_mean @pc1
print("x pca\n",x_pca) #projected on pc1
```

```
Covariance matrix:  [[0.44157656 0.40796209]
 [0.40796209 0.80668395]]
Eigenvalues:  [0.17718628 1.07107423]
Eigenvectors:  [[-0.83918077  0.54385259]
 [ 0.54385259  0.83918077]]
Eigenvalues:  [1.07107423 0.17718628]
Eigenvectors:  [[ 0.54385259 -0.83918077]
 [ 0.83918077  0.54385259]]
pc1
 [0.54385259 0.83918077]
x pca
 [ 1.21448825  0.8254791  -1.63952881  0.34590796  0.90825007  0.59122127
 -0.16364387 -1.50589016 -0.22499301 -0.69384358 -0.81170323 -0.73321207
  0.86480875 -0.45869648 -1.1340014  -0.72500845  0.61254057  1.91776654
 -1.53175086  0.24819983 -0.66663394  0.43233182  0.10916208 -0.22495449
 -0.79835893  1.82354689  2.13200742  1.55658637 -1.64722387 -1.33460733
 -0.83807642  0.27132571 -0.21498399  0.30966688  0.74718145 -0.321345
  0.08937017  1.17609254 -0.70488985 -1.37227624  1.1535438  -1.5855398
 -0.82297774  0.93996331  0.73747436 -0.00375135  0.7665174   0.57954938
 -0.10160133 -2.03844175 -0.40885573 -0.25505593  0.09423928  0.68608172
 -1.74118589 -1.36583974  1.84767718  1.47066203 -0.29797071  0.95327921
  1.88141465 -1.46805094 -0.6709691   1.7935376  -0.84072462 -0.93345665
 -1.15317951  0.3886115   0.73414924  1.25796171  0.20761144 -0.53847129
 -1.75593911  0.29574018 -1.46547922  0.06579856 -0.64011569  0.22254302
  1.20902717  0.26953849  0.22669422  0.44323761  0.31344911 -1.22860635
 -0.53019408 -1.06134862 -0.73758184 -0.64074089  0.09758807 -0.191742
 -1.41313655  2.04994159  0.09977612  0.85666314 -0.65673814  0.50841038
  1.2554567  -0.69537962  1.7275228   1.67910161]
```

```
import numpy as np
from sklearn.datasets import load_iris
iris = load_iris()
x = iris.data

# step 2 : Standardize
x_mean = x - np.mean(x,axis=0)

# step 3 : Covariance matrix
cov_matrix = np.cov(x_mean, rowvar=False)

# STEP 4 : Eigenvalues and Eigenvectors
eigenvalues, eigenvectors = np.linalg.eigh(cov_matrix)

#STEP 5 : sort eigenvectors by eigenvalues (descending)
sorted_idx = np.argsort(eigenvalues)[::-1]

eigenvalues = eigenvalues[sorted_idx]
eigenvectors = eigenvectors[:,sorted_idx]

#step 6 : project data onto first principal component (reduce to 1D)
pc1=eigenvectors[:,:2]
x_pca=x_mean @pc1

print("Eigenvalues: ",eigenvalues)
print("Principal components: ",eigenvectors)
print("x pca\n",x_pca) #projected on pc1
```

```
Eigenvalues:  [4.22824171 0.24267075 0.0782095  0.02383509]
Principal components:  [[-0.36138659  0.65658877  0.58202985  0.31548719]
 [ 0.08452251  0.73016143 -0.59791083 -0.3197231 ]
 [-0.85667061 -0.17337266 -0.07623608 -0.47983899]
 [-0.3582892  -0.07548102 -0.54583143  0.75365743]]
x pca
 [[ 2.68412563  0.31939725]
 [ 2.71414169 -0.17700123]
 [ 2.88899057 -0.14494943]
 [ 2.74534286 -0.31829898]
 [ 2.72871654  0.32675451]
 [ 2.28085963  0.74133045]
 [ 2.82053775 -0.08946138]
 [ 2.62614497  0.16338496]
 [ 2.88638273 -0.57831175]
 [ 2.6727558  -0.11377425]
 [ 2.50694709  0.6450689 ]
 [ 2.61275523  0.01472994]
 [ 2.78610927 -0.235112  ]
 [ 3.22380374 -0.51139459]
 [ 2.64475039  1.17876464]
 [ 2.38603903  1.33806233]
 [ 2.62352788  0.81067951]
```

```
[ 2.64829671  0.31184914]
[ 2.19982032  0.87283904]
[ 2.5879864   0.51356031]
[ 2.31025622  0.39134594]
[ 2.54370523  0.43299606]
[ 3.21593942  0.13346807]
[ 2.30273318  0.09870885]
[ 2.35575405 -0.03728186]
[ 2.50666891 -0.14601688]
[ 2.46882007  0.13095149]
[ 2.56231991  0.36771886]
[ 2.63953472  0.31203998]
[ 2.63198939 -0.19696122]
[ 2.58739848 -0.20431849]
[ 2.4099325   0.41092426]
[ 2.64886233  0.81336382]
[ 2.59873675  1.09314576]
[ 2.63692688 -0.12132235]
[ 2.86624165  0.06936447]
[ 2.62523805  0.59937002]
[ 2.80068412  0.26864374]
[ 2.98050204 -0.48795834]
[ 2.59000631  0.22904384]
[ 2.77010243  0.26352753]
[ 2.84936871 -0.94096057]
[ 2.99740655 -0.34192606]
[ 2.40561449  0.18887143]
[ 2.20948924  0.43666314]
[ 2.71445143 -0.2502082 ]
[ 2.53814826  0.50377114]
[ 2.83946217 -0.22794557]
[ 2.54308575  0.57941002]
[ 2.70335978  0.10770608]
[-1.28482569  0.68516047]
```